Week 4: Independent Study and Group Cases

Tashkent University of Technology | Linear Algebra I | Fall 2026

Matrix rank and matrix inverse

21 proposed problems with new numerical data. Student version: no solutions. Problems 1-5 are basic, 6-12 build technique, and 13-15 are challenges. Problems 16-18 add consolidation and an extension; 19-21 form the group project required by the Week 4 syllabus.

Suggested six-hour workload: individual practice on problems 1-18 (about 3 hours); group work on cases 19-21 (about 2 hours); checking and writing the final explanation (about 1 hour). Work in groups of 2-4 for the three cases. Submit shared calculations and a brief statement of each member's contribution. Each student should be able to explain every step.

Reading: Strang, supplied Introduction to Linear Algebra, 4th edition, §2.5 and introductory rank/pivot material in §3.3.

All vectors use column notation. Every problem starts on a new PDF page. Use the blank space for your work. Give exact fractions and verify proposed inverses by multiplication.

Problem 1: Ranks by inspection - basic

Find the rank of each matrix and give a reason:

$$Z = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix}.$$

Working space

Problem 2: Independent and dependent columns - basic

Let $A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 1 \\ 1 & 1 & 5 \end{bmatrix}$. Express column 3 as a combination of columns 1 and 2. Determine the rank and describe the plane containing every output $A\mathbf{x}$.

Working space

Problem 3: Rank by row reduction - basic

Reduce $C = \begin{bmatrix} 1 & 4 & 0 & 1 \\ 2 & 9 & 1 & 3 \\ 3 & 13 & 1 & 4 \end{bmatrix}$ to echelon form. State its rank and select independent columns from the original matrix.

Working space

Problem 4: Confirm rank with a minor - basic

For $A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 1 \\ 1 & 1 & 5 \end{bmatrix}$, use its determinant and a 2×2 minor to establish the exact rank. Why is the full determinant alone insufficient?

Working space

Problem 5: Diagonal inverse - basic

Find the inverse of $D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$. State what happens to its rank and inverse if its last diagonal entry is replaced by zero.

Working space

Problem 6: A 2 by 2 inverse - intermediate

Find A^{-1} for $A = \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$ using the determinant formula. Verify your answer by computing AA^{-1}.

Working space

Problem 7: Recover an input column - intermediate

Solve $A\mathbf{x} = \mathbf{b}$ using the inverse, where

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 13 \\ 3 \end{bmatrix}.$$

Check your solution in the original equation.

Working space

Problem 8: Gauss-Jordan with a row swap - intermediate

Use Gauss-Jordan elimination to invert $H = \begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$. Show the augmented matrix after the row swap and at the end.

Working space

Problem 9: Gauss-Jordan for a 3 by 3 matrix - intermediate

Find the inverse of $U = \begin{bmatrix} 1 & 4 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ by applying row operations to $[U \mid I_3]$. Verify the upper-right entry of UU^{-1}.

Working space

Problem 10: Inverse of a product - intermediate

Let $S = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Find $(SD)^{-1}$ without applying the 2×2 formula directly to SD. Check the order of the factors.

Working space

Problem 11: Prove that an inverse does not exist - intermediate

Let $B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$. Find a nonzero column $\mathbf{z}$ with $B\mathbf{z} = \mathbf{0}$. Use it to explain why B has no inverse. State its rank.

Working space

Problem 12: Rank and information loss in xyz space - intermediate

Let $P = \text{diag}(1, 1, 0)$ and $L = \text{diag}(1, 2, 4)$. Find their ranks. Compute the images of $\mathbf{a} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$ under P. Which transformation can be reversed?

Working space

Problem 13: Rank changes with a parameter - challenge

For

$$A(k) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & k-4 & 0 \\ 0 & 0 & k+1 \end{bmatrix},$$

find the rank for every real k. For which k does the inverse exist? Find that inverse when it exists.

Working space

Problem 14: A one-sided inverse - challenge

Let $A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Compute AB and BA. Does B qualify as an ordinary two-sided inverse of A? State the ranks of A and B.

Working space

Problem 15: Solve a matrix equation - challenge

Find the matrix X satisfying $AXB = C$, where

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 27 \\ 1 & 7 \end{bmatrix}.$$

Show which side you multiply on at each step.

Working space

Problem 16: Test an inverse claim - consolidation

A student claims that the inverse of $A = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ is $B = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/5 \end{bmatrix}$ and that $(A + I)^{-1} = B + I$. Check both claims by multiplication. Correct any false claim.

Working space

Problem 17: Transpose and inverse - consolidation

Let $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$. Compute $(A^T)^{-1}$ and $(A^{-1})^T$ independently. Verify their equality. Explain why the inverse of a symmetric invertible matrix is also symmetric.

Working space

Problem 18: Rank of a product can decrease - challenge

Let $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $Q = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Find the ranks of P, Q, and PQ. Explain why the rank of a product need not equal the smaller of the two factor ranks. Describe the action on $\begin{bmatrix} x \\ y \end{bmatrix}$.

Working space

Problem 19: Group case A: audit a rank calculation

Your group receives

$$C = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 1 \\ 3 & 7 & 4 & 1 \end{bmatrix}.$$

One report says its rank is 3 because all rows are nonzero; another says its rank is 2. Settle the disagreement by showing row operations. Select independent original columns and express each remaining column using them. Verify one nonzero 2×2 minor. Present the corrected reasoning in a short paragraph.

Working space

Problem 20: Group case B: construct and verify an inverse

For

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix},$$

compute the inverse by Gauss-Jordan elimination. Show the augmented matrix after each operation, then verify both products with your proposed inverse. Use it to solve $A\mathbf{x} = \begin{bmatrix} 2 \\ 6 \\ 7 \end{bmatrix}$. Have one group member check the result independently by substitution.

Working space

Problem 21: Group case C: can measured coordinates be recovered?

Two devices apply the transformations

$$L = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Determine each rank. Recover the input that gives $L\mathbf{x} = \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}$. Give two distinct inputs producing $P\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Sketch the image of a unit cube under each map on labeled xyz axes. Explain which information is lost and provide a short group conclusion.

Working space